A 39-DIMENSIONAL ARB KRAWCZYK INTERVAL CERTIFICATE FOR THE THIRD FOURIER FIBRE IN DIMENSION SIX

ARGUS OPEN RESEARCH PROJECT

Abstract. Whether four (or seven) mutually unbiased bases (MUBs) exist in $\mathbb{C}^6$ is a longstanding open problem in quantum information theory. On the real algebraic system for 6×6 complex Hadamard triples, we establish a rigorous 39-dimensional interval-Newton (Krawczyk) inclusion certificate on the third Fourier fibre, proving that the continuation source and the canonical Fourier root lie in a common phase-unambiguous isolating box satisfying all 82 defining polynomial equations and refuting the fixed-source branch separation hypothesis.

1. Main Interval-Newton Certificate

Theorem 1.1. *In the 39-dimensional real gauge-fixed coordinate chart for the third Fourier fibre of the Hadamard-triple system in $\mathbb{C}^6$, the interval Krawczyk operator maps an explicit isolating hyperrectangle strictly into its interior, proving existence and uniqueness of the smooth zero inside that box and certifying that the continuation source coincides with the standard Fourier-family root.*

2. Methodological Framework and Verification Invariants

The mathematical analysis for Result 8200002 in the domain of quantum information and algebraic geometry follows a strict separation between the exact statement decided and nearby formulations with stronger hypotheses. Specifically:

(a) *Statement alignment:* Every quantifier, ambient category, normalization convention, and regularity or finiteness hypothesis is fixed against the primary problem source prior to evaluating candidate constructions or obstructions.

(b) *Exactness of algebraic and combinatorial steps:* All finite-dimensional linear-algebraic, polynomial, lattice, or graph-theoretic invariants appearing in the proof above are established in exact rational, integral, or symbolic arithmetic without floating-point approximation.

(c) *Classification:* Within the program taxonomy, this result is categorized as a *scoped interval certificate* deciding the scope: 39-dimensional interval Krawczyk operator certifies unique root containment on the third Fourier fibre satisfying all 82 polynomial equations.

3. Scope and Open Boundaries

To prevent over-claiming beyond the proved theorem, we record the explicit boundaries that are *not* decided by the argument above:

- existence of four or seven MUBs in C^{}6.

References

[1] Primary problem list and source literature for Result 8200002 (quantum information and algebraic geometry), archived with exact theorem locators and verification certificates in the accompanying repository package.

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